

Teorija polj

Vektorski operatorji

$$\text{Gradient: } \nabla f = \left(\frac{\partial f}{\partial x}, \frac{\partial f}{\partial y}, \frac{\partial f}{\partial z} \right)$$

$$\text{Divergenca: } \nabla \cdot \mathbf{F} = \frac{\partial F_x}{\partial x} + \frac{\partial F_y}{\partial y} + \frac{\partial F_z}{\partial z}$$

$$\text{Rotor: } \nabla \times \mathbf{F} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ F_x & F_y & F_z \end{vmatrix}$$

$$\text{Laplacijan: } \nabla^2 f = \frac{\partial^2 f}{\partial x^2} + \frac{\partial^2 f}{\partial y^2} + \frac{\partial^2 f}{\partial z^2}$$

Lastnosti polj

$$\text{Irotacijalno: } \nabla \times \mathbf{F} = 0$$

$$\text{Solenoidno: } \nabla \cdot \mathbf{F} = 0$$

$$\text{Konservativno: } \mathbf{F} = -\nabla \phi$$

$$\text{Potencial: } \phi = - \int \mathbf{F} \cdot d\mathbf{r}$$

Prostorska geometrija

Koordinatni sistemi

$$\text{Polarni: } x = r \cos \theta, \quad y = r \sin \theta$$

$$\text{Cilindrični: } x = r \cos \theta, \quad y = r \sin \theta, \quad z = z$$

$$\text{Sferični: } x = r \sin \phi \cos \theta, \quad y = r \sin \phi \sin \theta, \quad z = r \cos \phi$$

Dodatno

$$\text{Ločna dolžina: } s = \int_a^b |\mathbf{r}'(t)| dt = \int_a^b \sqrt{x'(t)^2 + y'(t)^2 + z'(t)^2} dt$$

$$\text{Tangentni vektor: } \mathbf{T} = \frac{\mathbf{r}'(t)}{|\mathbf{r}'(t)|}$$

$$\text{Normalna ravnina: } \mathbf{r}'(t_0) \cdot (\mathbf{r} - \mathbf{r}(t_0)) = 0$$

$$\text{Odvod integrala s parametrom: } \frac{d}{dx} \int_a^b f(x, t) dt = \int_a^b \frac{\partial f}{\partial x} dt$$

$$\text{Naravna parameterizacija: } |\mathbf{r}'(s)| = 1, \quad s = \text{ločna dolžina}$$

Večkratni integrali

$$\text{Dvojni integral: } \iint_D f(x, y) dA = \int_a^b \int_{g_1(x)}^{g_2(x)} f(x, y) dy dx$$

$$\text{Ploščina: } A = \iint_D 1 dA$$

$$\text{Prostornina: } V = \iiint_D f(x, y) dA$$

$$\text{Težišče: } \bar{x} = \frac{1}{A} \iint_D x dA, \quad \bar{y} = \frac{1}{A} \iint_D y dA$$

$$\text{Trojni integral: } \iiint_E f(x, y, z) dV$$

$$\text{Masa: } m = \iiint_E \rho(x, y, z) dV$$

$$\text{Vztrajnostni moment: } I = \iiint_E r^2 \rho dV$$

$$\text{Povprečna vrednost: } \bar{f} = \frac{1}{V} \iiint_E f(x, y, z) dV$$

Krivoljnji integrali

Skalarno polje:

$$\int_C f ds = \int_a^b f(\mathbf{r}(t)) |\mathbf{r}'(t)| dt$$

Vektorsko polje:

$$\int_C \mathbf{V} \cdot d\mathbf{r} = \int_a^b \mathbf{V}(\mathbf{r}(t)) \cdot \mathbf{r}'(t) dt$$

Ploskovni integrali

Skalarno polje:

$$\iint_S f dS = \iint_D f(\mathbf{r}(u, v)) \sqrt{EG - F^2} du dv$$

$$\text{kjer } E = \mathbf{r}_u \cdot \mathbf{r}_u, \quad F = \mathbf{r}_u \cdot \mathbf{r}_v, \quad G = \mathbf{r}_v \cdot \mathbf{r}_v$$

Vektorsko polje:

$$\iint_S \mathbf{V} \cdot d\mathbf{S} = \iint_D [\mathbf{V}, \mathbf{r}_u, \mathbf{r}_v] du dv$$

Integralski izreki

$$\text{Greenov: } \oint_C (P dx + Q dy) = \iint_D \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} \right) dA$$

$$\text{Stokesov: } \oint_C \mathbf{V} \cdot d\mathbf{r} = \iint_S (\nabla \times \mathbf{V}) \cdot d\mathbf{S}$$

$$\text{Gaussov: } \iint_S \mathbf{V} \cdot d\mathbf{S} = \iiint_E \nabla \cdot \mathbf{V} dV$$

1 Complex Analysis (Kompleksna analiza)

1.1 Complex Functions

$$z = x + iy, \quad |z| = \sqrt{x^2 + y^2}, \quad \arg z = \arctan\left(\frac{y}{x}\right) \quad (1)$$

$$e^z = e^x (\cos y + i \sin y) \quad (2)$$

$$\log z = \ln |z| + i \arg z + 2\pi i k \quad (3)$$

$$\sin z = \frac{e^{iz} - e^{-iz}}{2i}, \quad \cos z = \frac{e^{iz} + e^{-iz}}{2} \quad (4)$$

$$\sinh z = \frac{e^z - e^{-z}}{2}, \quad \cosh z = \frac{e^z + e^{-z}}{2} \quad (5)$$

1.2 Analyticity

Cauchy-Riemann equations:

$$\frac{\partial u}{\partial x} = \frac{\partial v}{\partial y}, \quad \frac{\partial u}{\partial y} = -\frac{\partial v}{\partial x} \quad (6)$$

$$\text{Harmonic condition: } \nabla^2 u = \nabla^2 v = 0$$

1.3 Complex Integration

$$\text{Cauchy's formula: } f(z_0) = \frac{1}{2\pi i} \oint_C \frac{f(z)}{z - z_0} dz \quad (7)$$

$$\text{Residue theorem: } \oint_C f(z) dz = 2\pi i \sum \text{Res}(f, z_k) \quad (8)$$

$$\text{Laurent series: } f(z) = \sum_{n=-\infty}^{\infty} a_n (z - z_0)^n \quad (9)$$

Integrali

Pravila integriranja

Substitucija: $\int f(g(x))g'(x)dx = \int f(u)du, \quad u = g(x)$

Po delih: $\int u \, dv = uv - \int v \, du$

Parcialni ulomki: $\frac{P(x)}{Q(x)} = \sum \frac{A_i}{(x - a_i)^{n_i}}$

Elementarni integrali

$$\begin{aligned}\int x^n dx &= \frac{x^{n+1}}{n+1} + C, \quad n \neq -1 \\ \int \frac{1}{x} dx &= \ln|x| + C \\ \int e^x dx &= e^x + C \\ \int \sin x dx &= -\cos x + C \\ \int \cos x dx &= \sin x + C \\ \int \frac{1}{1+x^2} dx &= \arctan x + C \\ \int \frac{1}{\sqrt{1-x^2}} dx &= \arcsin x + C\end{aligned}$$

Odvodi

Pravila odvajanja

$$\begin{aligned}(f+g)' &= f' + g' \\ (fg)' &= f'g + fg' \\ \left(\frac{f}{g}\right)' &= \frac{f'g - fg'}{g^2} \\ (f(g(x)))' &= f'(g(x)) \cdot g'(x)\end{aligned}$$

Elementarni odvodi

$$\begin{aligned}\frac{d}{dx} x^n &= nx^{n-1} \\ \frac{d}{dx} e^x &= e^x \\ \frac{d}{dx} \ln x &= \frac{1}{x} \\ \frac{d}{dx} \sin x &= \cos x \\ \frac{d}{dx} \cos x &= -\sin x \\ \frac{d}{dx} \tan x &= \sec^2 x \\ \frac{d}{dx} \arctan x &= \frac{1}{1+x^2} \\ \frac{d}{dx} \arcsin x &= \frac{1}{\sqrt{1-x^2}}\end{aligned}$$